

Cybersecurity

Introduction to Linux

Lesson 3.1



Class Objectives

By the end of class, you will be able to:

- 1 Name three of the most important distributions of Linux.
- 2 Navigate the Linux file structure by using the command line.
- 3 Discuss the prominence and utility of the command line for IT and security.
- 4 Use commands for basic file navigation and manipulation.
- 5 Navigate deeply nested folder structures using file paths.



So, Why **Linux**?

Why Linux

Millions of websites run a version of Linux.



Why Linux

The ubiquity of Linux machines on modern networks makes it a common target for attackers.





Familiarity with the
operating system is
crucial for cybersecurity
professionals.



Linux in a Professional Context

Knowledge is essential for the following technical roles, among others:



Helpdesk/IT support



System administration



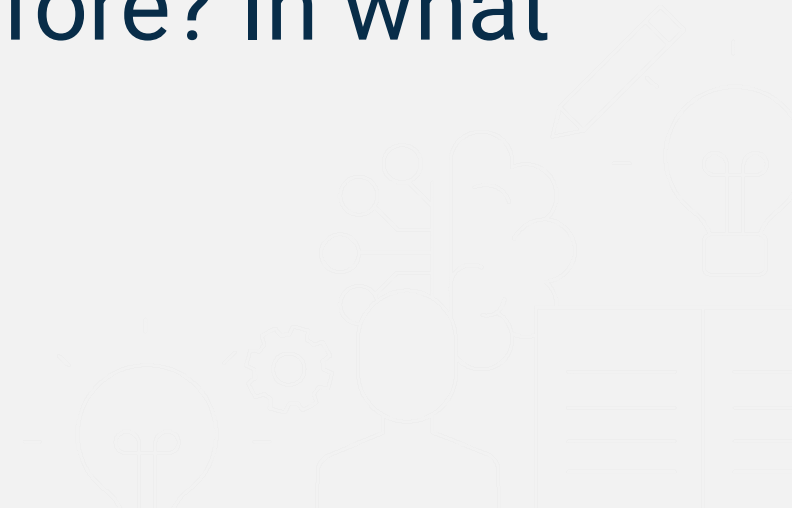
Penetration testing



Network forensics



Have you ever used
Linux before? In what
context?



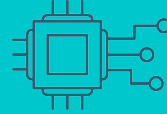
Common Linux Applications



**Web servers
and websites**



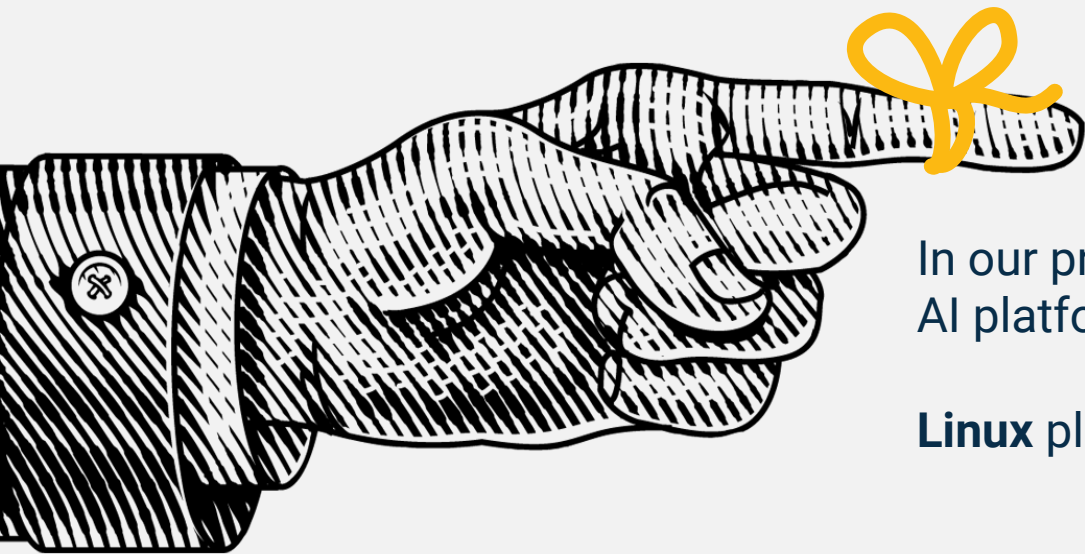
Smartphones



**Embedded
systems**



**Government
and military
systems**



Remember,

In our previous class, we discussed various AI platforms and technologies.

Linux plays a pivotal role in the AI landscape.





Linux History and **Distributions**

A (very) Brief History of Linux

01

In 1969–1970, Dennis Ritchie, Kenneth Thompson, and several others at AT&T Bell Labs began developing a small operating system called Unix.

02

Within the next two decades, different updates and distributions of Unix were created. Most were proprietary.

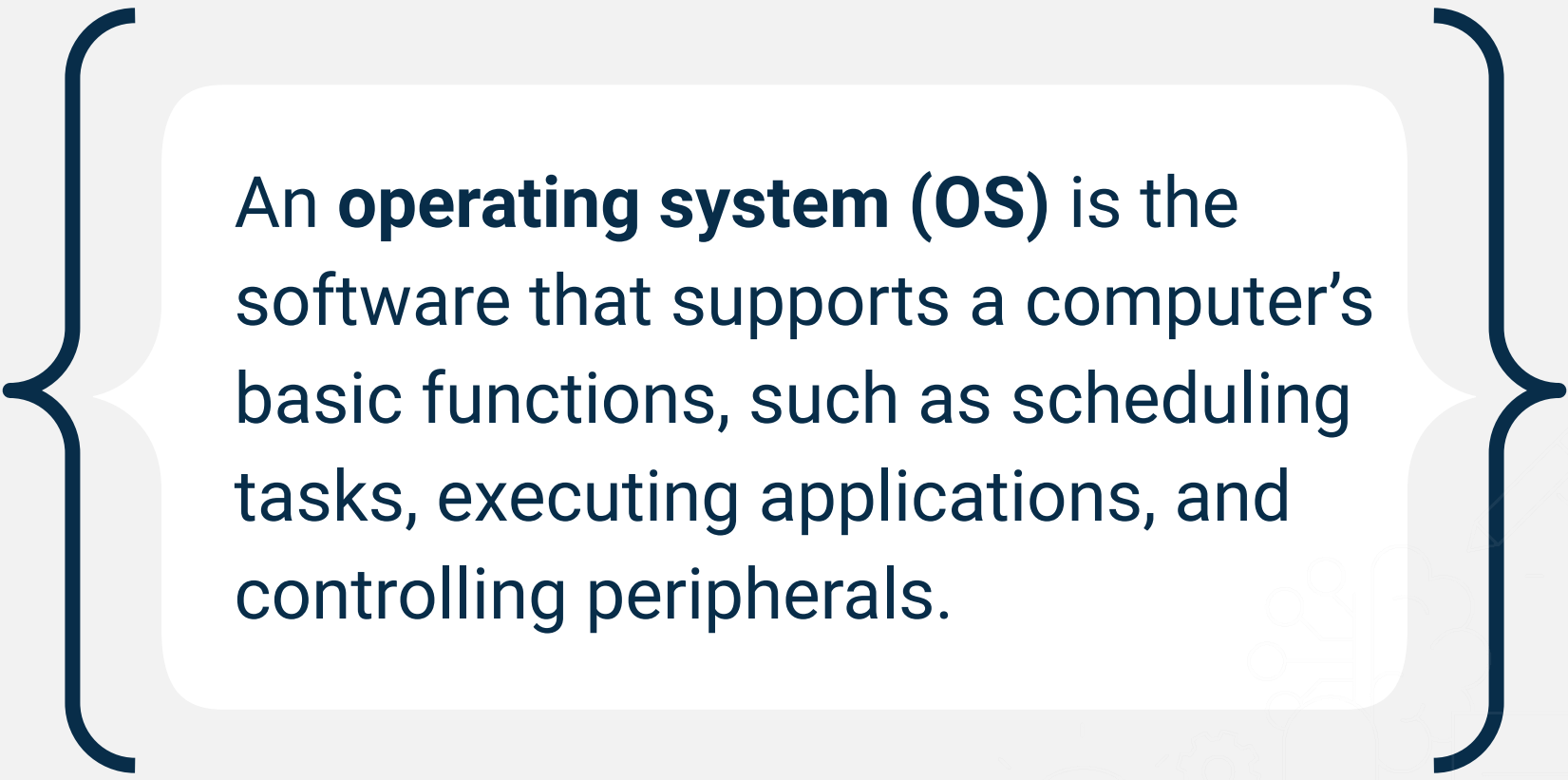
A (very) Brief History of Linux

03

In 1984, Richard Stallman's Free Software Foundation (FSF) started the GNU project, which created a free distribution of the Unix operating system.

04

In 1991, Linus Torvalds used many of the components from the GNU project to create a free, open source operating system, now known as Linux.

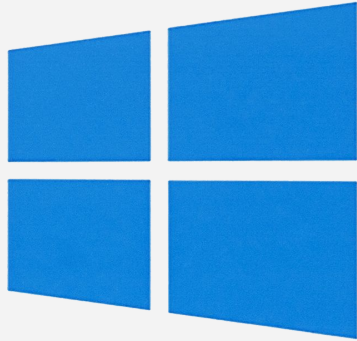


An **operating system (OS)** is the software that supports a computer's basic functions, such as scheduling tasks, executing applications, and controlling peripherals.

Examples of Operating Systems (OS)

01

Windows



02

macOS



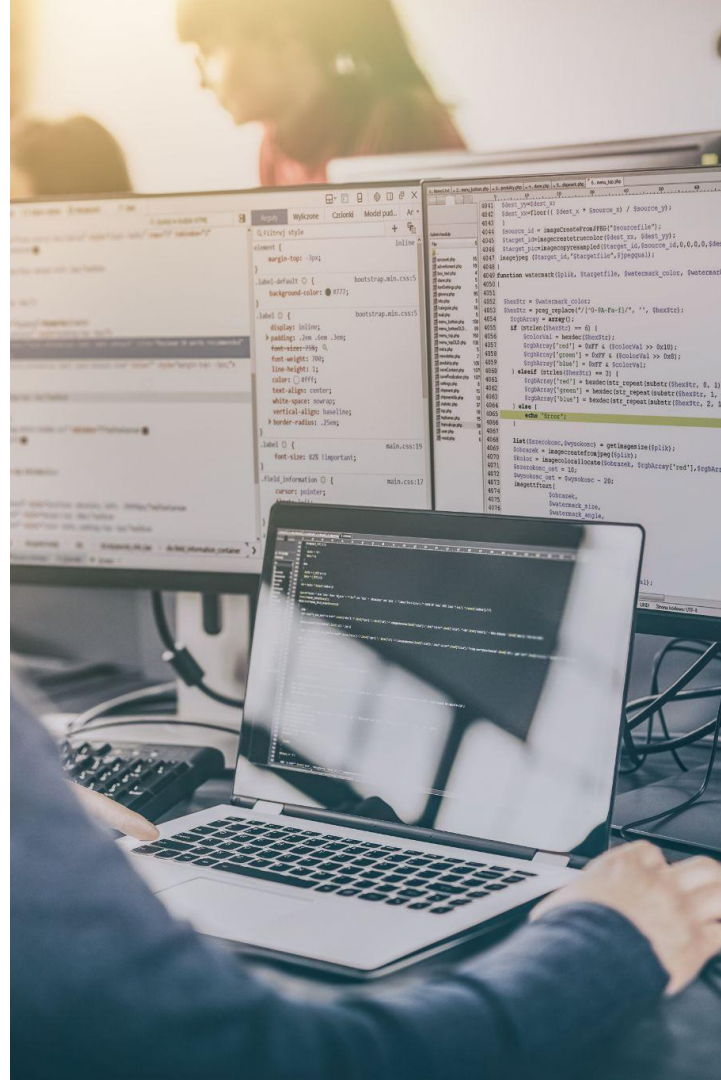
03

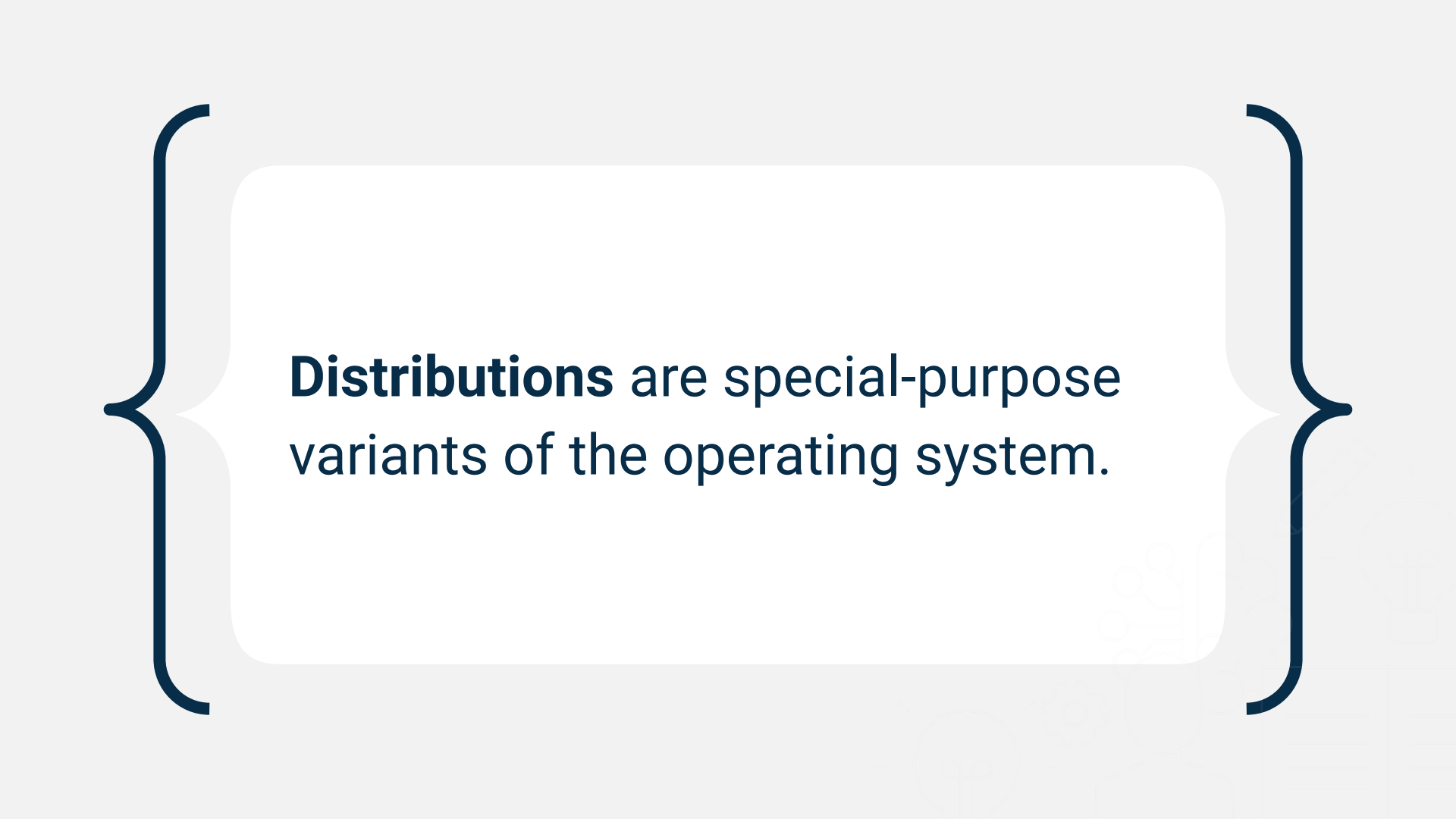
Linux



Why Linux?

- Unlike Windows and OS X, Linux is **Free, Open Source Software (FOSS)**.
 - **Anyone** can modify the source code.
- Most tools in the hacker/security community are still published as open source.
- Free software and information is prevalent in the industry.
 - This includes almost all the tools we'll use in class.





Distributions are special-purpose variants of the operating system.

Linux Distributions

In this course, we'll use two distributions: Ubuntu and Kali Linux.

01

Ubuntu

Ubuntu is geared towards general-purpose users.



02

Kali Linux

Kali is designed specifically for security professionals.



Linux Distributions

Ubuntu and Kali Linux are both specialized distributions of **Debian**, which is itself a distribution.



Command Line

In order to conserve as many resources as possible, most production Linux servers don't offer a graphical interface.

Therefore knowledge of the command line is essential for understanding Linux.



Command line-only machines are called **headless servers**.

Benefits of headless servers:

- Require very few resources
- Give the server maximum resources to run its services and applications
- Do not require GUI-based system





Activity:

Distribution Research

You are a system administrator at X Corp, which has recently experienced a number of breaches involving servers running outdated Linux distributions.

- In response, the IT Department has decided to upgrade the affected servers with newer distributions of Linux.
- You must conduct research to determine which distribution is most appropriate for each machine.

Suggested Time:
15 Minutes

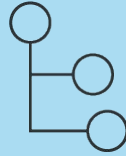


Time's up!
Let's review



Questions?





Linux File **System** Structure

Linux File System Structure



Conventional file system structure



Consistent conventions across distributions

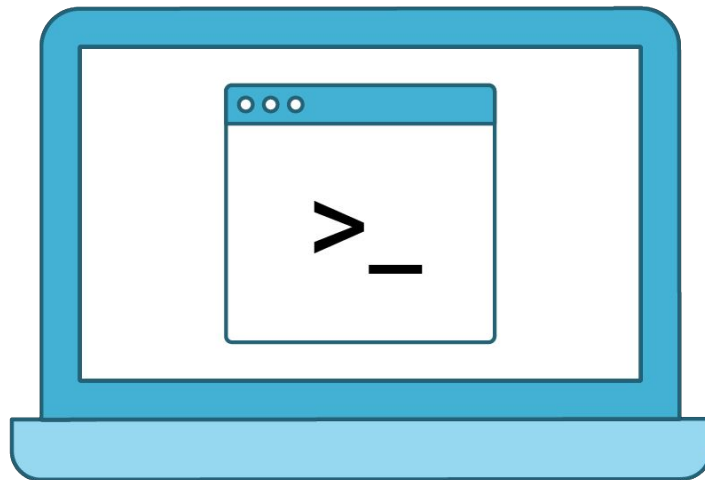


Mostly uses bash shell by default

Root

Root can have multiple meanings in Linux.
Depending on the context, root can be either:

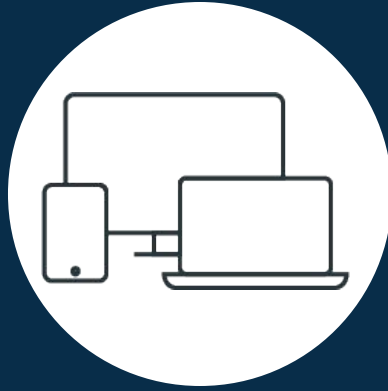
- The user account for the administrator
- The root directory that contains every other directory
- The home directory for the root user



File Systems

Linux maintains certain conventions for storing files:

<code>/</code>	Contains every other directory.
<code>/home</code>	Contains users' private files. Users should not be able to save files elsewhere.
<code>/root</code>	The home directory of the root user.
<code>/etc</code>	Contains configuration files, defining how a machine runs and who can use it.
<code>/bin, /sbin</code>	Contains main binary or program files.
<code>/var</code>	Contains files that change over time.
<code>/tmp</code>	Contains files that are only needed for a short period of time.



Instructor **Demonstration**

Tour Linux File System

Demo Takeaways

The audit took us to the following directories:

/etc	Stores specific, system-wide configuration files, as well as the most sensitive files on a Linux system
/var	Stores files that are continually updated
/home	Stores user home folders
/tmp	Where applications write temporary files that can be deleted on reboot
/bin and /sbin	Where the system keeps its main binary or program files



Why **Terminal** Matters

Graphical User Interfaces

We are used to GUIs with their familiar icons, windows, buttons, and mouse pointers.

Mac



PC





Fist to Five

Raise a Fist: If you've never worked with the command line and barely know what it is

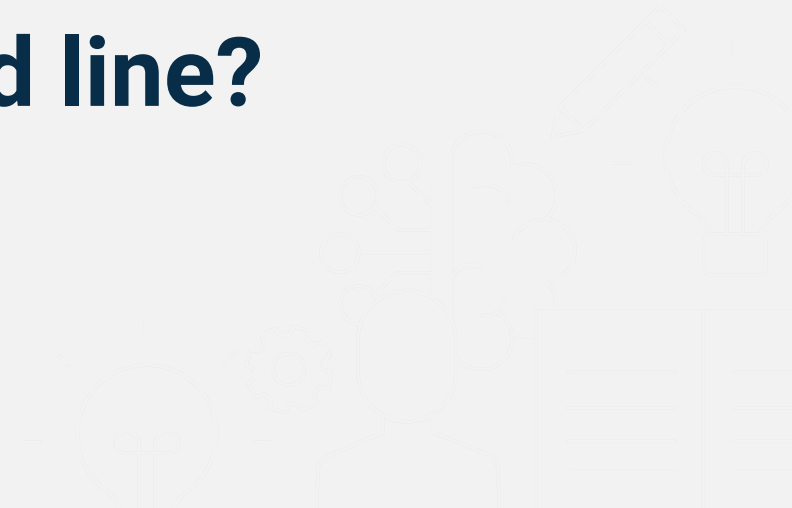
Raise a Five: If you work with the command line on a daily basis

Raise One, Two, Three, or Four: If you fall somewhere in between





What is the
command line?



What Is the Command Line?

Command line users **don't** directly interact with icons and buttons on the screen.

Users complete tasks on their computers by issuing commands in the terminal.



Why the Command Line?

Using the command line is a critical skill. At different times, it may be:

01

The only way to
achieve a desired
outcome

02

The fastest way to
achieve a desired
outcome

03

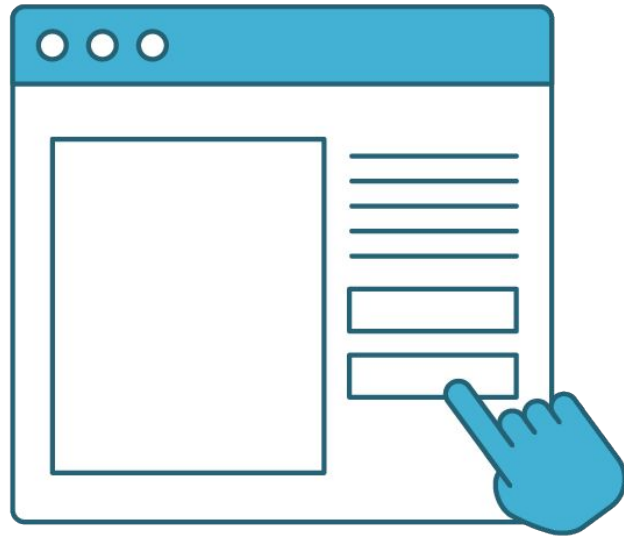
The most flexible
way to achieve a
desired outcome

Why the Command Line?

Using the command line is a critical skill. At different times, it may be:

01 The only way to achieve a desired outcome

You may have to work with systems and tools that have no GUI interface.



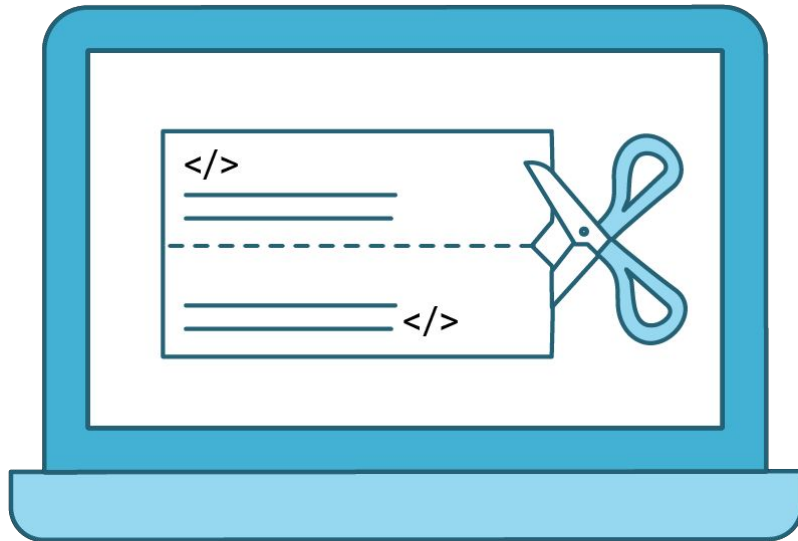
Why the Command Line?

Using the command line is a critical skill. At different times, it may be:

02

The fastest way to achieve
a desired outcome

The command line has tools that can speed
up a task.



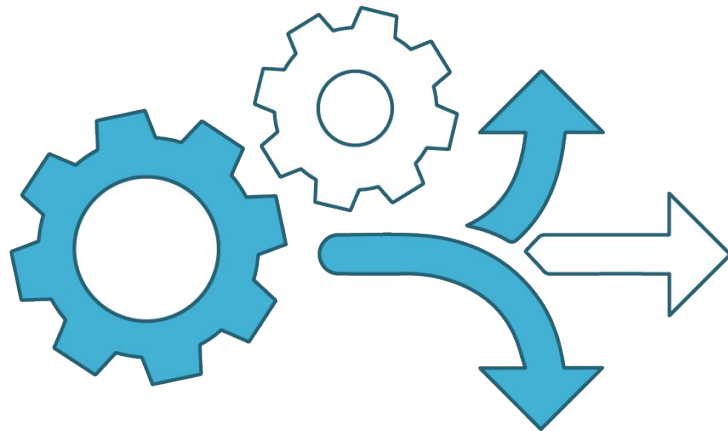
Why the Command Line?

Using the command line is a critical skill. At different times, it may be:

03

The most flexible way to achieve a desired outcome

Working directly with the command line provides greater control.



Words to the Wise

A few things to remember



The best way to learn the command line is to use it.



This section is heavily focused on practical activities.



Take notes during each demo and try your best on activities.



Remember: Practice makes perfect!



Activity Scenario

Welcome to Candy Corp

- ❖ You are a security analysts at a candy manufacturing company called Candy Corp.
- ❖ You will be using the command line to investigate a potential rogue employee who may be selling secret Candy Corp recipes to a rival organization, Sugar Corp.
- ❖ You will be tasked with using the command line to identify evidence on the system of suspicious activity from this rogue employee



Questions?



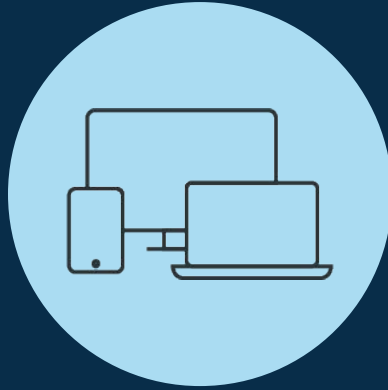


Basic **Terminal** Commands

Basic Commands

In the next demo, we'll cover the following text commands:

<code>pwd</code>	Displays the current working directory ("print working directory")
<code>ls</code>	Lists directories and files in the current directory
<code>cd</code>	Navigates into a directory ("change directory")
<code>mkdir</code>	Makes a directory
<code>rmdir</code>	Removes a directory
<code>touch</code>	Creates an empty file
<code>rm</code>	Removes a file
<code>clear</code>	Clears the terminal history of a page
<code>echo</code>	Displays text on the terminal
<code>history</code>	Displays a list of commands previously run



Instructor Demonstration

Common Terminal Commands

Demonstration Review

What does each of the commands do?

<code>pwd</code>	Displays the current working directory ("print working directory")
<code>ls</code>	Lists directories and files in the current directory
<code>cd</code>	Navigates into a directory ("change directory")
<code>mkdir</code>	Makes a directory
<code>rmdir</code>	Removes a directory
<code>touch</code>	Creates an empty file
<code>rm</code>	Removes a file
<code>clear</code>	Clears the terminal history of a page
<code>echo</code>	Displays text on the terminal
<code>history</code>	Displays a list of command previously ran



Activity:

Take Five and Practice the Command Line

In this activity, you are a security analyst at Candy Corp.

- Candy Corp believes an employee is selling secret recipes.
- You must create evidence directories for email, logs, and web access to organize your investigation.

Suggested Time:
20 Minutes





Time's up!
Let's review

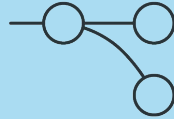


Questions?





Break
15 mins



Relative vs. Absolute **Paths**



File Paths

A file path is a unique location in the file system where a file is located.

The `cat.jpg` image file is located in the `my_images` directory, which is in the `Desktop` directory, in the `Sally` directory, in the `home` directory

```
/home/Sally/Desktop/my_images/cat.jpg
```

The directory path is
`/home/Sally/Desktop/my_images/`

The file is an image
called
`cat.jpg`



File Paths in IT

You are a security administrator and need to access a network log for an investigation.

You might ask the network team for the location or path of the file.

The network team would provide you with a file path such as:

```
/var/log/logfile1.txt
```

File Paths

There are two types of file paths:

01

Absolute paths

An **absolute path** indicates the location of a file regardless of your current location in the file system.

```
/home/Sally/Desktop/Sallysfiles/textfiles/Sallyfile.txt
```

02

Relative paths

A **relative path** starts at your current location in the file system and doesn't require using the root directory ("/").

```
Sallysfiles/textfiles/Sallyfile.txt
```



Absolute Paths

We want to access a file located at:

```
/home/Sally/Desktop/Sallysfiles/textfiles/Sallyfile.txt
```

```
/home/bob/file_logs/
```

```
/home/Sally/Desktop/Sallysfiles/textfiles/Sallyfile.txt
```



Relative Paths

We can utilize the dotslash

- We can utilize ./ to indicate a relative path from our current location:
 - . indicates current directory
 - / start of the relative path

If we are working in the directory:

```
/home/Sally/Desktop
```

```
./Sallysfiles/textfiles/Sallyfile.txt
```

File Paths on the Command Line

Relative and absolute paths function as helpful shortcuts on the command line.

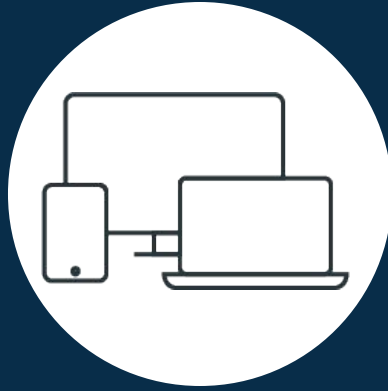
```
cd Desktop
```

```
cd Sallysfiles
```

```
cd textfiles
```



```
cd Desktop/Sallysfiles/textfiles/
```



Instructor **Demonstration**

Relative and absolute paths



Demonstration Scenario

Our manager at ACME Corp needs us to create several additional evidence files for our investigation. They provide the following instructions:

1 Navigate to the directory `/03-instructor/day1/pathnav_demonstration/security_evidence/Case2/`. Create an empty file called `case2_evidence`.

2 Navigate to the directory `Case1` and create an empty file called `Web_logs`.

3 Navigate back to the `home` folder.



Copying and Moving **Files**



Security and IT professionals are often asked to **copy** files from one location to another.

Copying Files

The command “**cp**” creates a copy of the file.



The original file is located in the directory.



The **cp** (copy) command is entered on the terminal.



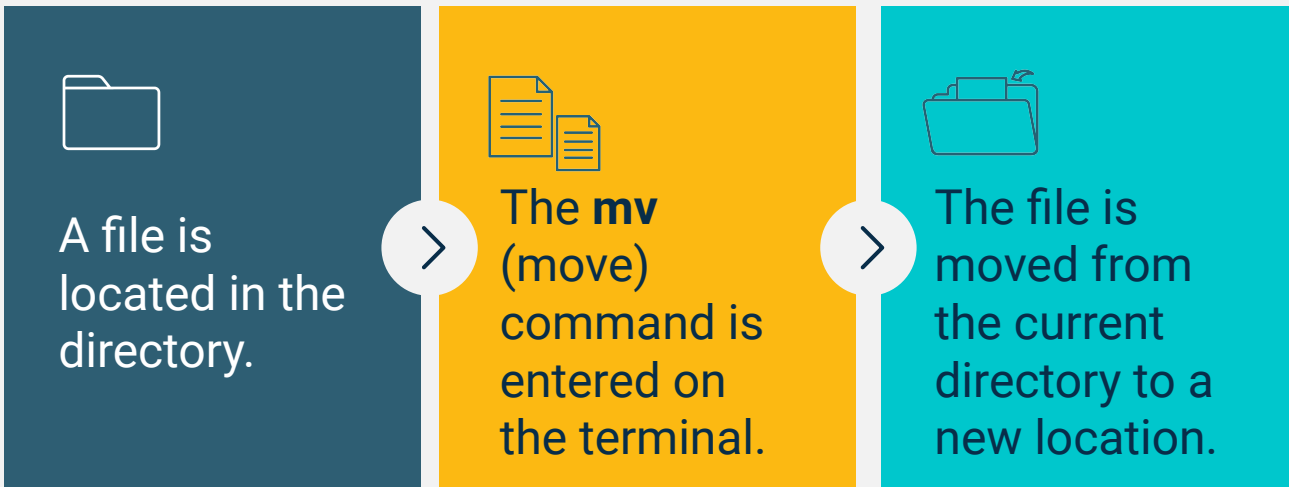
The original file remains in current directory.



A copy is placed in new location.

Moving Files

The command “**mv**” moves the file from the current location into a new location.



Remember: Unlike **cp**, the original file doesn't remain.

Similar Syntax

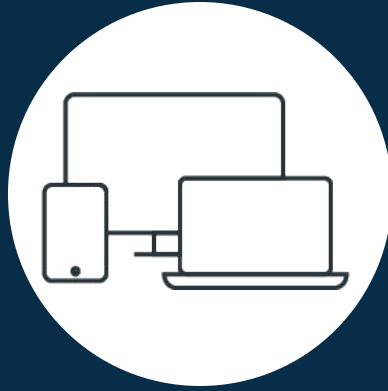
```
<command> <file> <location to copy or move>
```

```
cp cat.jpg /dirA
```

- This command copies the file `cat.jpg` into the directory **dirA**.
- The **<file>** or **<location>** can be indicated with an absolute or relative path.

```
mv dirA/dirB/dirC/cat.jpg  
/dirA/dirB
```

This command moves the file **cat.jpg** from the **dirC** directory over to the **dirB** directory.



Instructor **Demonstration**

Copy and Move Files

Let's Review

Let's review the following concepts and commands:

File path	Identifies a unique location in a file system
Absolute path	Starts at the top directory of the file system
Relative path	Begins from the current directory
cp	Copies files
mv	Moves files



Activity:

Find your Milky Way

Continue in your role as a security analyst at Candy Corp.

- Your manager at Candy Corp needs you to create an additional directory, as they believe there is a second employee working with Sugar Corp.
- You must also copy and move several of the evidence files after creating this new directory.

Suggested Time:
20 Minutes



Time's up!
Let's review



Questions?





Recap

In today's class you:

01

Discussed Linux distributions

02

Navigated Linux file structure using command line

03

Described utility of command line in security professions

04

Performed basic file manipulation

05

Discussed relative and absolute file paths



Thank you